

CHAPTER 8 : POLLUTION

Topic:-

Visit the square having heavy traffic in your area and report pollution at different times of day.

Introduction:-

Air pollution is one of the greatest environmental and health challenges facing modern cities and towns. The burning of fossil fuels (petrol and diesel) in millions of cars, auto-rickshaws, motorbikes, and heavy trucks releases hazardous gases and microscopic soot particles into the air. In our cities, busy road intersections and traffic squares (chowks) are hotspots where vehicular pollution is concentrated.

Vehicles idling at red traffic signals discharge toxic gases like Carbon Monoxide (CO), Nitrogen Oxides (NO_x), Sulfur Dioxide (SO₂), and Particulate Matter (PM_{2.5} and PM₁₀). Continuous exposure to vehicular smog causes eye irritation, coughing, chronic bronchitis, asthma, and reduced lung capacity in children. For our Class 8 science project on Pollution, we conducted a field survey at the busiest traffic square in our locality to observe, measure, and report air and noise pollution across different times of the day.

❖ Main Pollutants Released by Vehicles:-

- **Carbon Monoxide (CO):** Colorless poison gas that reduces oxygen in blood.
- **Particulate Matter (PM_{2.5}):** Microscopic soot particles that penetrate deep into lungs.
- **Lead & Unburnt Hydrocarbons:** Toxic exhaust causing headaches and dizziness.
- **Noise Pollution:** Constant loud honking (exceeding 85 decibels) causing hearing damage.

Aim & Survey Plan:-

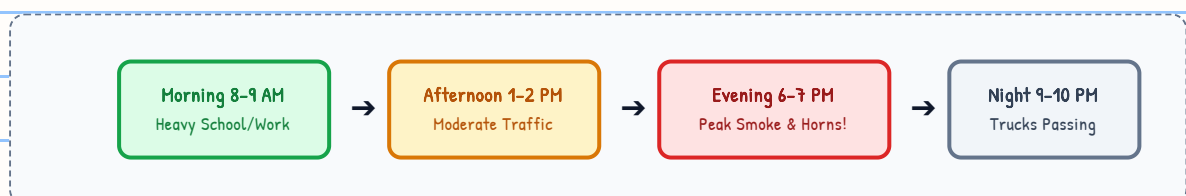
Aim: To survey a major traffic square, count vehicle density, monitor air quality (AQI) and noise levels across four different time slots, and report findings with recommendations.

❖ Objectives of the Field Survey:-

- To record vehicle counts during morning rush, afternoon lull, evening rush, and night.
- To note physical symptoms experienced by pedestrians and traffic police constables.
- To use mobile apps (AQI India and Sound Level Meter) to gather real-time data.

❖ Survey Location & Tools:-

- **Location:** Shivaji Chowk (main central 4-way intersection in our city).
- **Tools Used:** Field clipboard, stopwatch, smartphone with sound meter app, N95 face masks.



Survey Methodology & Data Collection:-

Our student team of 4 members followed strict safety rules while surveying Shivaji Chowk:

- **Safe Observation Post:** We stood safely on the elevated pedestrian sidewalk next to the traffic booth, wearing reflective jackets and N95 masks.
- **15-Minute Vehicle Sampling:** During each of the 4 time slots, we ran a 15-minute stopwatch and counted two-wheelers, auto-rickshaws, cars, and heavy buses/trucks.
- **Noise Level Measurement:** We held the phone with the Sound Level Meter app and recorded the average decibel (dB) readings over 10 minutes.
- **Visible Smoke & Dust Log:** We noted haze levels, soot deposits on roadside ficus tree leaves, and interviewed the traffic police constable on duty.

❖ Physical Symptoms Noted by Student Observers:-

After just 45 minutes of standing at the intersection during evening peak hours, two students experienced burning sensation in eyes and throat dryness!



Traffic Square Pollution Report Table:-

The table below summarizes our field survey observations across the four time periods:

Time Slot	Vehicles / 15 Min	Average Noise (dB)	Air Quality (AQI)	Visible Condition & Smell
Morning (8:00–9:00 AM)	520 vehicles	82 dB (Very loud)	168 (Moderate)	Mild smoke haze; rush of school buses & bikes
Afternoon (1:00–2:00 PM)	210 vehicles	68 dB (Normal)	112 (Moderate)	Clearer air; bright sun disperses exhaust
Evening (6:00–7:30 PM)	780 vehicles	94 dB (Harmful!)	245 (Poor / Unhealthy)	Thick black smog; choking petrol fumes; non-stop horns
Night (9:30–10:30 PM)	290 vehicles	74 dB (Loud)	185 (Moderate)	Heavy diesel trucks; dark soot particles visible

Scientific Analysis & Discussion:-

Analyzing our field data revealed critical environmental facts about urban traffic:

- **Peak Pollution Hours:** Evening rush hour (6:00 to 7:30 PM) showed the highest pollution: AQI soared to 245 and noise hit 94 dB! This happens because commercial vehicles, office commuters, and shoppers converge simultaneously, creating bumper-to-bumper traffic jams.
- **Idling Engines Waste Fuel:** When cars wait at a 90-second red light with engines running, fuel burns incompletely, releasing 3 times more poisonous Carbon Monoxide (CO) than when moving smoothly.
- **Impact on Roadside Plant Life:** We wiped leaves of roadside banyan and neem trees. A thick black crust of carbon soot coated every leaf, blocking stomata pores and choking plant photosynthesis.
- **Health Toll on Traffic Police:** The on-duty constable shared that traffic police officers suffer from chronic throat coughs, tinnitus (ringing in ears), and eye fatigue due to 8-hour daily exposure.

Actionable Solutions & Government Measures:-

To reduce dangerous intersection pollution, we recommend several proven measures:

- **1. 'Switch Off at Red Light' Campaign:** Turning off vehicle ignition at signals with countdown timers longer than 20 seconds saves fuel and cuts toxic exhaust by 40%.
- **2. Promotion of Electric Vehicles (EVs):** Switching city auto-rickshaws and two-wheelers to electric batteries produces zero tailpipe emissions and runs silently.
- **3. Strict PUC Enforcement:** Vehicles without valid Pollution Under Control (PUC) certificates should face heavy fines.
- **4. Green Noise Barriers:** Planting dense rows of broad-leaved trees (Neem, Peepal, Ashoka) along road dividers absorbs noise and filters airborne soot particles.

❖ No-Honking Zone Rule:-

Honking should be strictly banned near schools, hospitals, and residential squares with automated sound-activated traffic cameras.

What We Learned & Conclusion:-

- Traffic squares are major urban pollution hotspots where vehicle exhaust and honking concentrate.
- Evening rush hours experience dangerous AQI levels (>240) and noise levels exceeding 90 dB.
- Vehicle idling at red lights wastes fuel and causes heavy carbon soot deposition on roadside trees.
- Switching to public transport, electric vehicles, and cycling reduces urban smog effectively.

Conclusion:-

Our field study of Shivaji Chowk provided clear, firsthand proof of how vehicular traffic deteriorates urban air and sound quality. By documenting pollution spikes across different hours, we realized that curbing traffic pollution requires both personal discipline (switching off engines, carpooling) and progressive municipal urban planning (electric buses, cycle tracks). Every citizen must act to keep our city skies clean and breathable.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 8: Pollution).
- Maharashtra Pollution Control Board (MPCB) Urban Ambient Air Quality Reports.
- Central Pollution Control Board (CPCB) National Ambient Noise Monitoring Network.